

9. (a) Use a hyperbolic substitution and calculus to show that

$$\int \frac{x^2}{\sqrt{x^2 - 1}} dx = \frac{1}{2} \left[x\sqrt{x^2 - 1} + \operatorname{arcosh} x \right] + k$$

where k is an arbitrary constant.

(6)

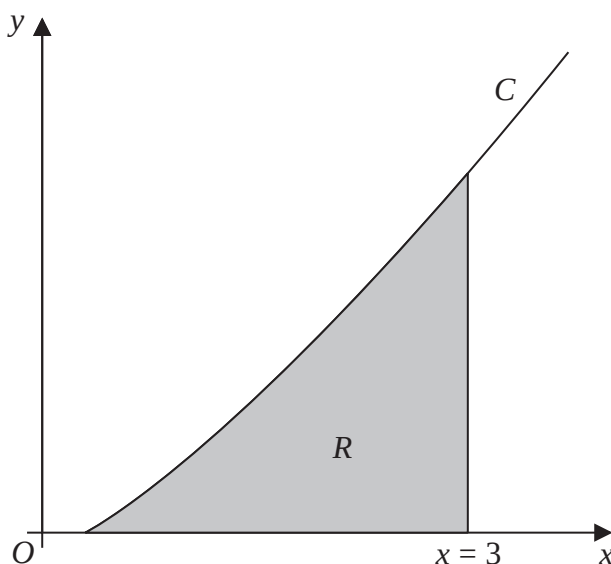


Figure 1

Figure 1 shows a sketch of part of the curve C with equation

$$y = \frac{4}{15} x \operatorname{arcosh} x \quad x \geq 1$$

The finite region R , shown shaded in Figure 1, is bounded by the curve C , the x -axis and the line with equation $x = 3$

- (b) Using algebraic integration and the result from part (a), show that the area of R is given by

$$\frac{1}{15} \left[17 \ln(3 + 2\sqrt{2}) - 6\sqrt{2} \right]$$

(5)



2.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Determine the values of x for which

$$64 \cosh^4 x - 64 \cosh^2 x - 9 = 0$$

Give your answers in the form $q \ln 2$ where q is rational and in simplest form.

(4)

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9. (i) (a) Explain why $\int_0^{\infty} \cosh x \, dx$ is an improper integral.

(1)

(b) Show that $\int_0^{\infty} \cosh x \, dx$ is divergent.

(3)

(ii) $4 \sinh x = p \cosh x$ where p is a real constant

Given that this equation has real solutions, determine the range of possible values for p

(2)

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2. (a) Write $x^2 + 4x - 5$ in the form $(x + p)^2 + q$ where p and q are integers.

(1)

(b) Hence use a standard integral from the formula book to find

$$\int \frac{1}{\sqrt{x^2 + 4x - 5}} dx$$

(2)

(c) Determine the mean value of the function

$$f(x) = \frac{1}{\sqrt{x^2 + 4x - 5}} \quad 3 \leq x \leq 13$$

giving your answer in the form $A \ln B$ where A and B are constants in simplest form.

(3)

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1. (a) Using the definition of $\sinh x$ in terms of exponentials, prove that

$$4\sinh^3 x + 3\sinh x \equiv \sinh 3x \quad (2)$$

- (b) Hence solve the equation

$$\sinh 3x = 19 \sinh x$$

giving your answers as simplified natural logarithms where appropriate. (5)



2.

$$f(x) = \tanh^{-1}\left(\frac{3-x}{6+x}\right) \quad |x| < \frac{3}{2}$$

(a) Show that

$$f'(x) = -\frac{1}{2x+3} \quad (4)$$

(b) Hence determine $f''(x)$ (1)(c) Hence show that the Maclaurin series for $f(x)$, up to and including the term in x^2 , is

$$\ln p + qx + rx^2$$

where p , q and r are constants to be determined. (3)



Answer ALL questions. Write your answers in the spaces provided.

1. (a) Prove that

$$\tanh^{-1}(x) = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) \quad -k < x < k$$

stating the value of the constant k .

(5)

- (b) Hence, or otherwise, solve the equation

$$2x = \tanh\left(\ln\sqrt{2-3x}\right)$$

(5)

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1. The curve C has equation

$$y = 31 \sinh x - 2 \sinh 2x \quad x \in \mathbb{R}$$

Determine, in terms of natural logarithms, the exact x coordinates of the stationary points of C .

(7)

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$$y = \cosh^n x \quad n \geq 5$$

(a) (i) Show that

$$\frac{d^2 y}{dx^2} = n^2 \cosh^n x - n(n-1) \cosh^{n-2} x \quad (4)$$

(ii) Determine an expression for $\frac{d^4 y}{dx^4}$

(2)

(b) Hence determine the first three non-zero terms of the Maclaurin series for y , giving each coefficient in simplest form.

(2)



1.

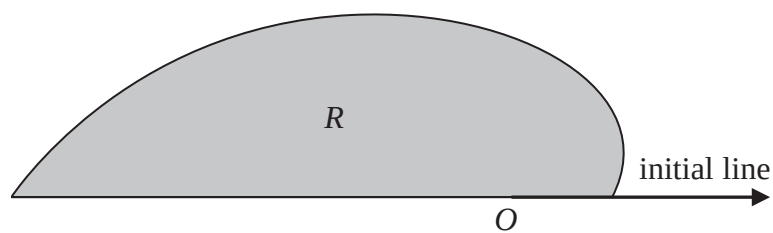
**Figure 1**

Figure 1 shows a sketch of the curve with polar equation

$$r = 2\sqrt{\sinh \theta + \cosh \theta} \quad 0 \leq \theta \leq \pi$$

The region R , shown shaded in Figure 1, is bounded by the initial line, the curve and the line with equation $\theta = \pi$

Use algebraic integration to determine the exact area of R giving your answer in the form $pe^q - r$ where p , q and r are real numbers to be found.

(4)

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6. Given that

$$y = e^{2x} \sinh x$$

prove by induction that for $n \in \mathbb{N}$

$$\frac{d^n y}{dx^n} = e^{2x} \left(\frac{3^n + 1}{2} \sinh x + \frac{3^n - 1}{2} \cosh x \right)$$

(6)

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